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Enhancing Decision-Making in Extended Reach Coiled Tubing Operations Through Reliable Telemetry and Downhole Tool Integration

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Abstract

This paper evaluates the use of real-time telemetry systems in coiled tubing (CT) operations to improve performance in extended reach and multilateral wells. The objective is to reduce downhole uncertainty, such as unknown Weight on Bottom Hole Assembly, Tool Pressure, and Depth location by integrating downhole sensors with reliable surface-to-downhole data transmission. These technologies are key for navigating complex extended well paths and improving decision-making during interventions.

The study focuses on conductor-based telemetry systems that use insulated wire inside CT to send data and power between downhole tools and the surface. Case studies include jobs using extended reach tools (ERT), hydraulic multilateral tools (HMLT), and sensors such as gamma ray (GR), casing collar locator (CCL), tension, and pressure gauges.

These systems were tested in high-vibration and long-reach environments. The setup enabled accurate depth tracking, lateral entry, and real-time response to downhole events, such as sticking or pressure spikes, through live monitoring of tool behavior and formation interaction.

Field and Technical results show that combining telemetry with advanced downhole tools improves CT operations in complex wells. In extended reach wells, the telemetry system reliably transmitted data even when ERTs were creating strong vibrations. This helped maintain continuous tool feedback and supported depth tracking using CCL and GR tools. In multilateral wells, telemetry-enabled HMLTs allowed operators to accurately enter specific lateral branches, reducing risk of wrong entry or tool deviation. Real-time readings of tension and pressure allowed early detection of downhole problems like pipe sticking or fluid losses, enabling quick adjustments. Across several jobs, this setup reduced non-productive time and improved the success rate of interventions, especially in long horizontal sections and branched wells. The system also improved confidence in tool positioning, making it easier to perform tasks like stimulation, cleanout, and re-entry. These benefits show that reliable telemetry can extend the reach of CT operations and help teams manage difficult well geometries more safely and effectively.

This paper presents new field-based evidence on how real-time telemetry, combined with mechanical and sensor tools, improves CT performance in complex well environments. The approach provides better control, safer operations, and improved results in extended reach and multilateral interventions offering

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a practical upgrade to traditional CT systems and filling a key gap in current well intervention practices. extended reach and multilateral interventions, making them invaluable in all oil and gas environments.

Introduction

Extended reach and multilateral wells are essential in modern reservoir development, enabling greater reservoir to contact and hydrocarbon recovery. However, they introduce increased complexity in well intervention, particularly when using Coiled Tubing (CT). Common issues in such operations include lack of real-time downhole data, poor depth tracking, and tool misplacement. These challenges are magnified in deviated or branched well geometries, increasing the risk of unsuccessful interventions and tool failures.

Traditionally, CT operations rely on surface indications and memory tools that cannot provide real-time feedback. This limitation affects operational control, often delaying responses to mechanical resistance, sticking, fluid losses, or tool deviation. To overcome this, conductor-based telemetry systems have been developed. These systems embed an insulated conductor cable inside the CT string, enabling continuous bi-directional communication between surface equipment and downhole tools.

This paper presents field-based evidence on the application of real-time telemetry-enabled CT systems in complex well environments, with a focus on Extended Reach Tools (ERT) and Hydraulic Multilateral Tools (HMLT). By integrating downhole sensors—such as gamma ray (GR), casing collar locator (CCL), tension, and pressure gauges—with reliable telemetry, operators gain real-time insight into tool behavior and formation interaction.

In extended reach wells, ERTs benefit from continuous data transmission despite high vibration environments, supporting accurate depth tracking and tool control. In multilateral wells, telemetry-enabled HMLTs allow precise lateral entry, reducing the risk of misruns and tool deviation. These capabilities enhance operational control, reduce non-productive time, and improve the overall success rate of CT interventions.

The results demonstrate that combining real-time telemetry with advanced mechanical tools significantly extends the reach and effectiveness of CT operations, offering a practical upgrade to conventional systems and addressing key limitations in current well intervention practices.

Technology Background

The methodology implemented in this study relies on the deployment of an advanced coiled tubing telemetry system that integrates a mono-conductor cable pre-installed within the coiled tubing string. This embedded conductor enables simultaneous power delivery and bidirectional data transmission between surface equipment and a downhole sensor assembly (BHA). The system was applied in extended reach wells requiring real-time monitoring of downhole parameters to support critical decision-making during milling, stimulation, and cleanout operations.

System Configuration and Deployment

The telemetry-enabled CT system consists of three key components:

Mono-Conductor Installation

A single insulated conductor is installed longitudinally within the coiled tubing. The conductor terminates in a quick-connect mechanical and electrical interface at the bottom of the CT.

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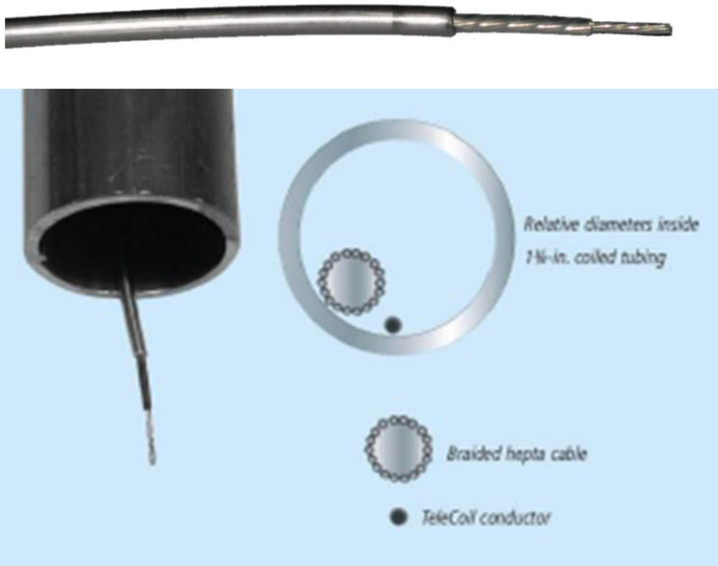

Figure 1—Mono-conductor

Downhole Sensor Assembly

The BHA includes sensors for internal/external pressure, temperature, and casing collar locator (CCL). Additionally, the assembly is equipped with standard CT functionalities such as double flapper check valves, a ball-activated wireline release, and dual-circulation subs to maintain operational integrity.

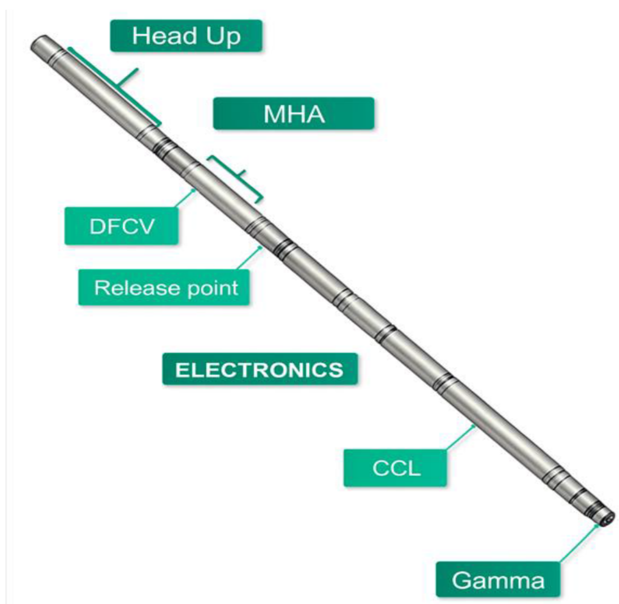

Figure 2—Downhole Sensor Assembly

Surface Acquisition and Processing Unit

At surface, the conductor is connected to a dedicated data acquisition system capable of real-time visualization and logging. This system interprets telemetry data and provides immediate feedback on downhole conditions.

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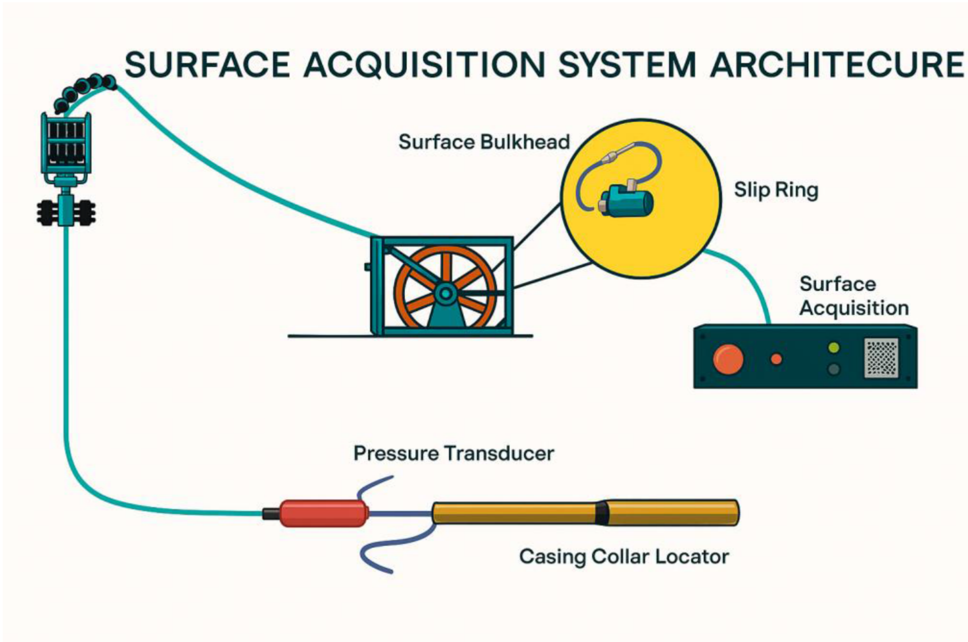

Figure 3—Surface Acquisition architecture

Extended Reach Tool (ERT)

The Extended Reach Tool (ERT) is a hydraulically powered friction reduction device installed at the end of the coiled tubing (CT string). Its function is to minimize friction and drag forces encountered as the CT is deployed deeper into the wellbore, particularly in extended reach or highly deviated wells.

The ERT converts hydraulic energy from the circulating fluid into axial oscillations along the CT. These oscillations reduce static friction by transforming it into lower dynamic friction, improving CT mobility. Additionally, the vibration lifts the CT off the low side of the wellbore, further decreasing contact forces and enhancing weight transfer to the bottom-hole assembly (BHA). This also generates an internal axial load toward the BHA, improving tool performance.

The ERT consists of two major components, power section and valve sections. The fluid is pumped through the power section which as a resultant change the orientation of valve sections. Valve section is further composed of two segments. One valve is attached to the rotor of the power section and other valve is attached to the bottom sub of the tool. Once the fluid is pumped through the tool, one valve changes its position against the other, while changing the total flow area during this whole process. Due to change in total flow area, periodic oscillations are created in the entire system that will be travelling along the coiled tubing string to reduce the friction between CT and wellbore. The extended reach tool used in this job was a 2-1/8-inch vibratory tool.

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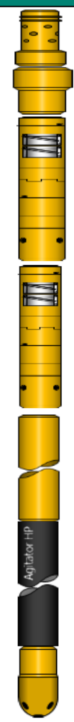
BHA PROPOSED for extended reach applications



OD (in)		Tool	Length (in)	Description
2.00			4.50	2" x 0.175" Internal Connector
2.25			6.00	NOGO Crossover
2.13			112.00	INTEGRATED SENSOR 3/4" Wire Release 5/8" BHA Release 1/2" Pass Thru
2.13			37.00	2 1/8" Motor Head Assembly 5/8" Disconnect Ball Seat , 1/2" Circulation Sub Ball Seat 7.5K Burst Disc
2.13			60.00	Straight Bar
2.13			63.60	ERT
2.13			7.00	Jetting Nozzle
		Total Length of BHA:	290.1	24.2

Figure 4—Bottom hole configuration for ERT.

Case Studies

In Well A, a high-complexity acid stimulation was performed in a horizontal well with a total measured depth (MD) of 19,101 ft and a 6-1/8″ open-hole section extending from 15,192 to 19,101 ft (3,909 ft), completed with 4-1/2″ production tubing to 5,036 ft, a 7″ liner, and open-hole from 6-1/8″ onward. The production tubing was equipped with an Electric Submersible Pump (ESP) and a Y-tool with an internal diameter of 2.441". The well exhibited a maximum deviation of 91.65° at 16,619 ft and a maximum dogleg severity of 5.1°/100 ft at 10,072 ft, presenting significant challenges for conveyance. The stimulation plans involved 2″ coiled tubing (CT) deployed with a 2-1/8″ Extended Reached Tools rated at 2,250 lbs of pulling force.

Three coiled tubing (CT) intervention runs were conducted in Well A to address conveyance and treatment challenges in an extended-reach horizontal well. Each run utilized a tailored bottom-hole assembly (BHA), equipped with downhole sensors that provided real-time measurements of differential pressure. These telemetry readings were critical for identifying mechanical lockups, confirming tool functionality (e.g., burst disk and tractor activation), and diagnosing failures such as not proper ETR functioning. Sensor feedback enabled dynamic adjustments to pumping rates, depth, speed, and tool deployment strategies, significantly improving operational efficiency and reducing non-productive time (NPT). The integration of sensor diagnostics into each run supported informed decision-making and enhanced reliability in complex well conditions.

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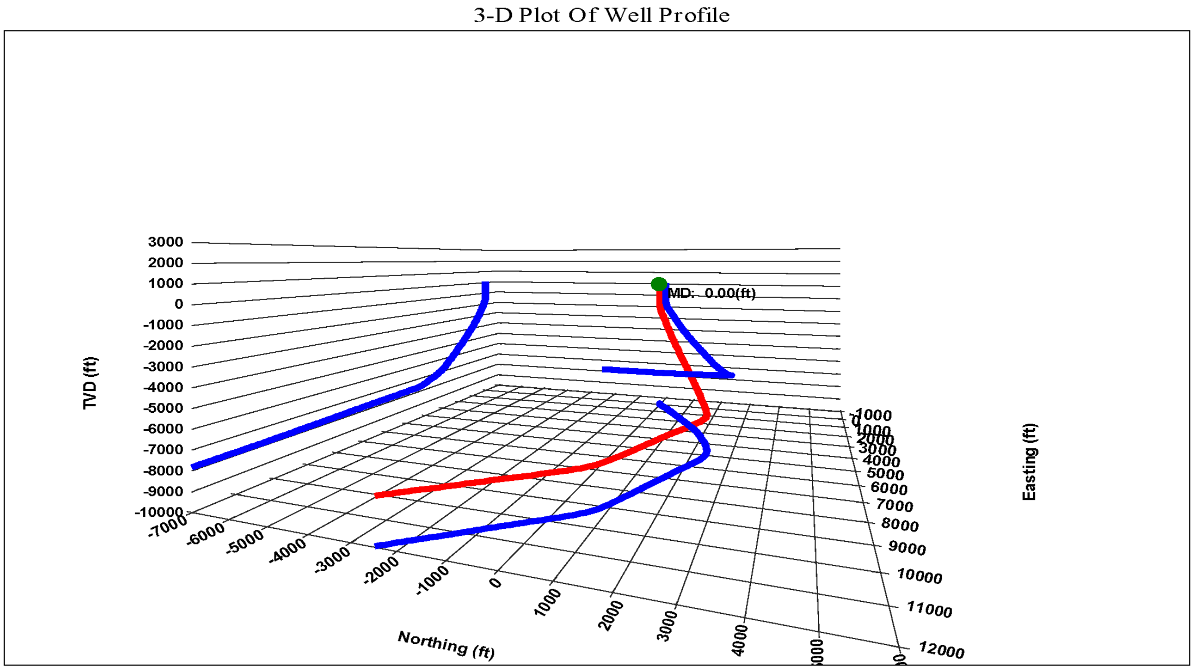

Figure 5—3D plot of well profile

RUN 1 – Well A Burst disk activation

A coiled tubing (CT) run was initiated using a conventional MHA with a 7,500-psi burst disk and a Vibration extended reach tool. CT was run in hole (RIH). Friction reducer pill was spotted at 5,189 ft. across the liner section. At 15,200 ft, a pre-flush was initiated at the nozzle, and pumping was increased to 1.2 bpm with CT to activate the Vibration extended reach tool. CT RIH continued into the open hole while pumping pre-flush until a mechanical lockup occurred at 17,457 ft, indicated by a 3,000-lbf weight drop and reduced on downhole telemetry tool pressure sensor. CT was picked up by 40 ft and RIH resumed; however, another lockup occurred at a shallower depth. After observing a sudden drop in tool differential pressure from 2,400 psi to 1,600 psi, the burst disk was confirmed to have activated as can see in Figure 6. A decision was made to pull out of hole (POOH) for tool inspection. Surface testing confirmed the burst disk had activated below the designed set pressure.

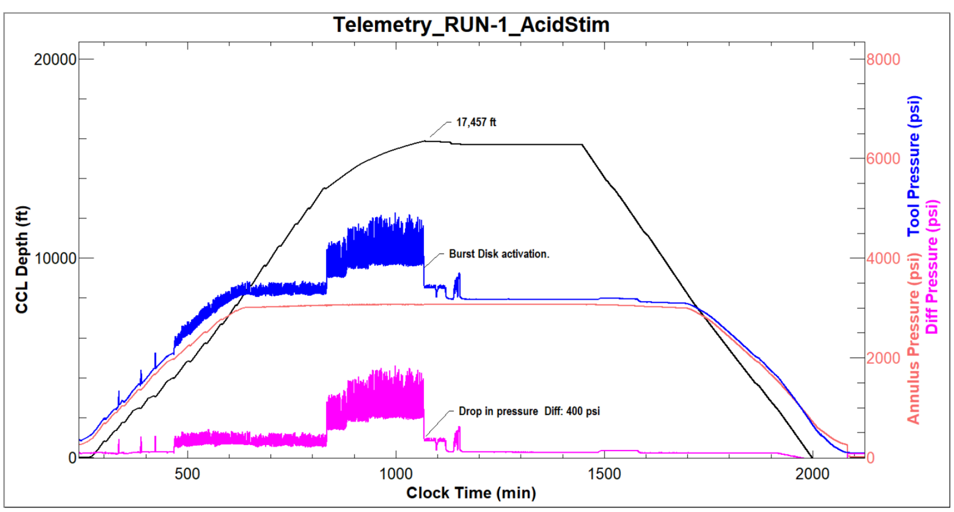

Figure 6—Down hole Telemetry reading for Run 1

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Run 2 – Well A Malfunction Hydraulic tractor

During a second intervention run in Well A, a hydraulic tractor tool was deployed to assist with acid stimulation in the extended horizontal section. The operation began by run-in-hole (RIH) procedures with frequent pull tests. At 5,100 ft, spotting of friction reducer lubricant commenced and continued to the liner shoe at 15,200 ft. At 9,800 ft, a slack weight test confirmed tractor functionality, evidenced by weight fluctuation and increased differential pressure, notice at Figure 7.

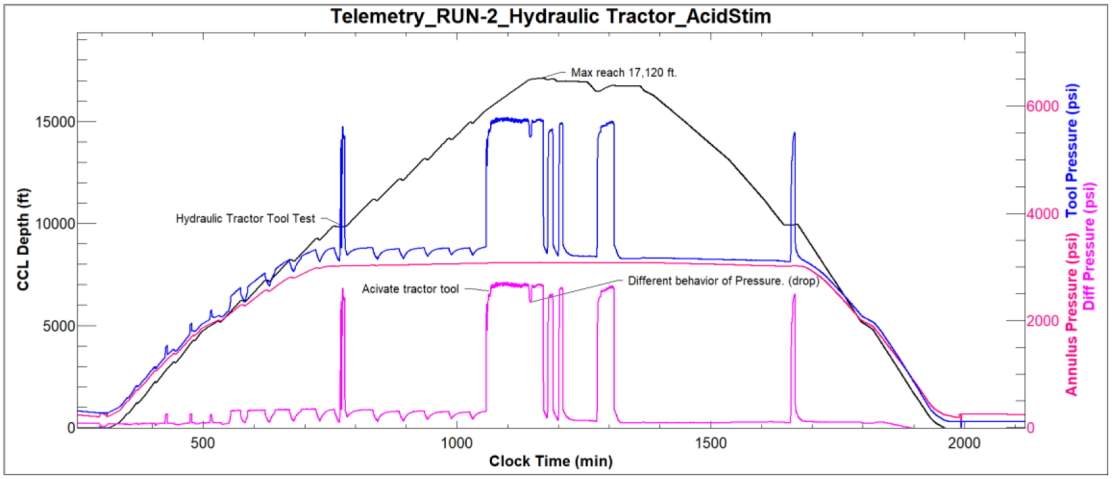

Figure 7—Down hole Telemetry reading for Run 2

Upon reaching 15,200 ft, a pre-flush was pumped at 1.4 bpm to activate the tractor. However, at 17,210 ft, the coiled tubing (CT) experienced a sudden 5,000 lb as can see at Figure 8. weight drop, prompting a stop in pumping and pull-out-of-hole (POOH) for inspection.

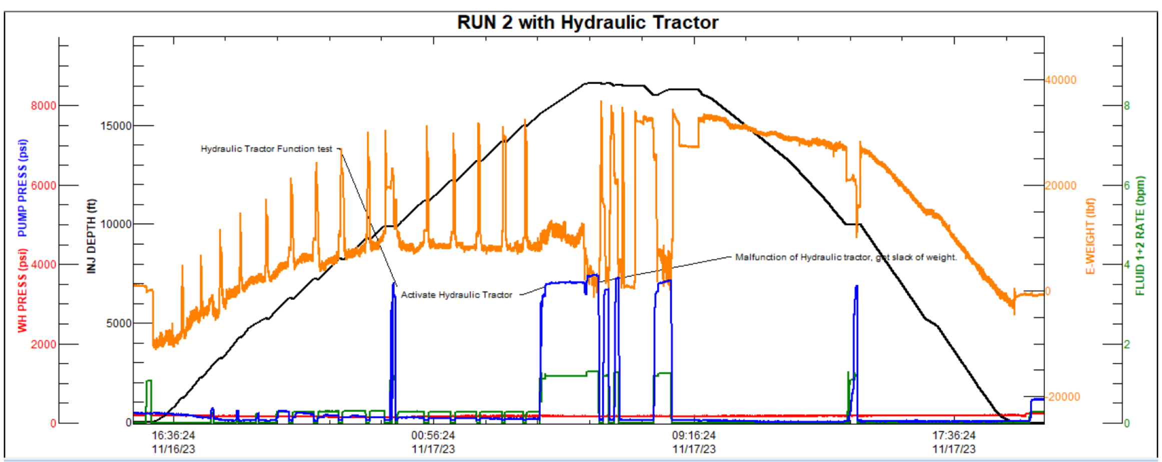

Figure 8—Run 2 overview

The weight normalized after picking up 30 ft, and RIH resumed, though further lockups were encountered. A POOH was executed, and a surface function test revealed that the bottom static gripper pins had sheared, rendering the tractor inoperable. The decision was made to terminate the run.

Run 3 – Well A Successful reached TD

During the third intervention run, a mechanical vibration tool was deployed to enhance conveyance and enable access to the target total depth (TD) in the horizontal open-hole section. A friction-reducing chemical

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pill was spotted across the interval from 5,100 ft to the 15,200 ft liner shoe to minimize drag forces. Upon reaching 15,200 ft, a pre-flush treatment was pumped at 1.4 bbl/min to activate the mechanical vibration tool functionality.

Coiled tubing (CT) was subsequently conveyed to a measured depth of 18,900 ft, corresponding to 95% of the planned open-hole interval. At this point, 3,000 lbf of slack weight was observed, indicating an obstruction or lockup condition. To initiate treatment, the circulation port in the bottomhole assembly (BHA) was opened to isolate the mechanical vibration tool and enable fluid placement, at Figure 10, shows the insolation of the tool.

A 20-stage acid stimulation treatment was executed while pulling out of hole (POOH) to 15,200 ft to maximize coverage of the treated interval. Following this, a run-in-hole (RIH) operation was conducted to deploy a post-flush treatment as reference in Figure 9. During the RIH, a mechanical lockup occurred at 17,257 ft, accompanied by a 4,000 lbf weight drop. A minor pickup and subsequent slack-off allowed the CT to reach 17,267 ft before a second lockup was encountered. This depth was designated as the operational TD for the run. The remaining volume of the post-flush treatment was placed at this final depth prior to POOH.

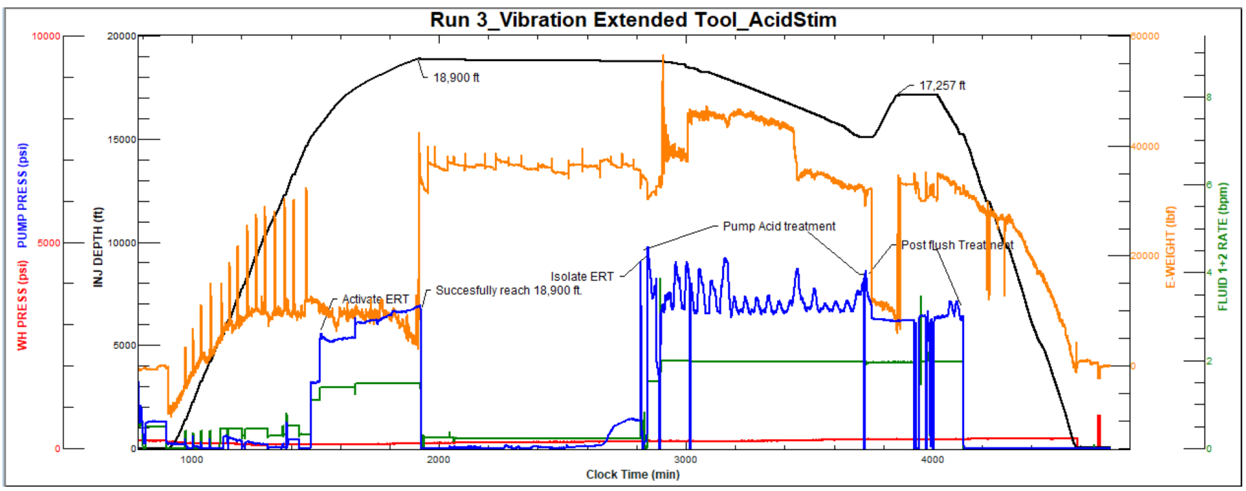

Figure 9—Run 3 overview shows successfully reach well TD and treatment pumped POOH

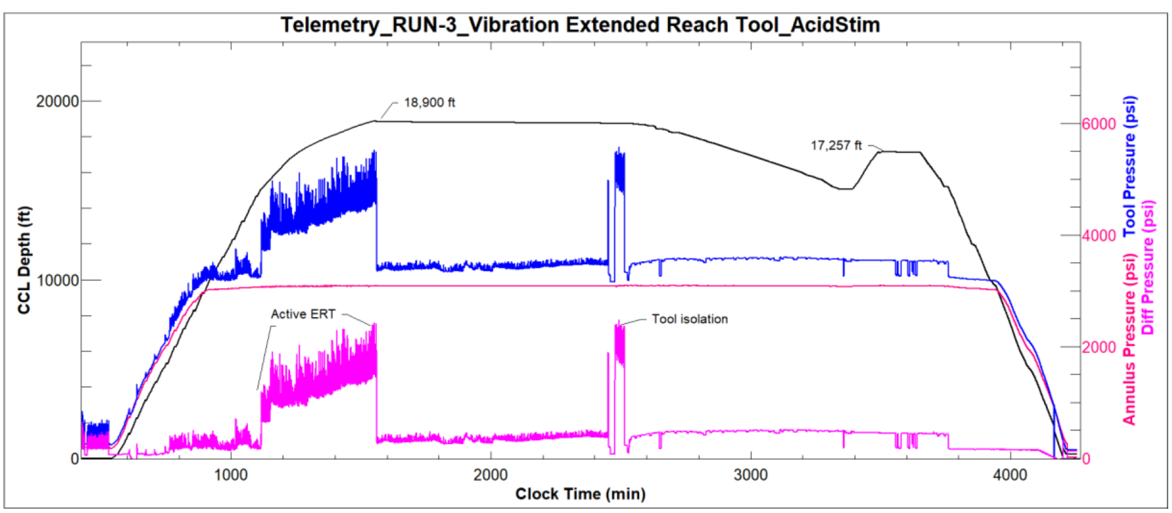

Figure 10—Down hole Telemetry readings for Run 2

In Well B, a sour oil producer, multiple intervention challenges were encountered, including a long, highly deviated 6-1/8″ open hole interval (17,687 to 20,713 ft MD) and a high hydrogen sulfide (H_2S)

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concentration of 15.23 mole%. To improve reservoir productivity, an extended-reach acid stimulation was planned. However, operational constraints—including a minimum internal restriction of 2.441″ at 5,692 ft MD and a maximum dogleg severity of 4.44°/100 ft—introduced substantial technical risk. As a result, the stimulation strategy was engineered to include multiple coiled tubing (CT) runs, each necessitated by different challenges encountered during execution. Each run utilized optimized bottomhole assemblies (BHAs) and specially formulated fluid systems to progressively access and treat the target interval without compromising wellbore integrity or operational safety.

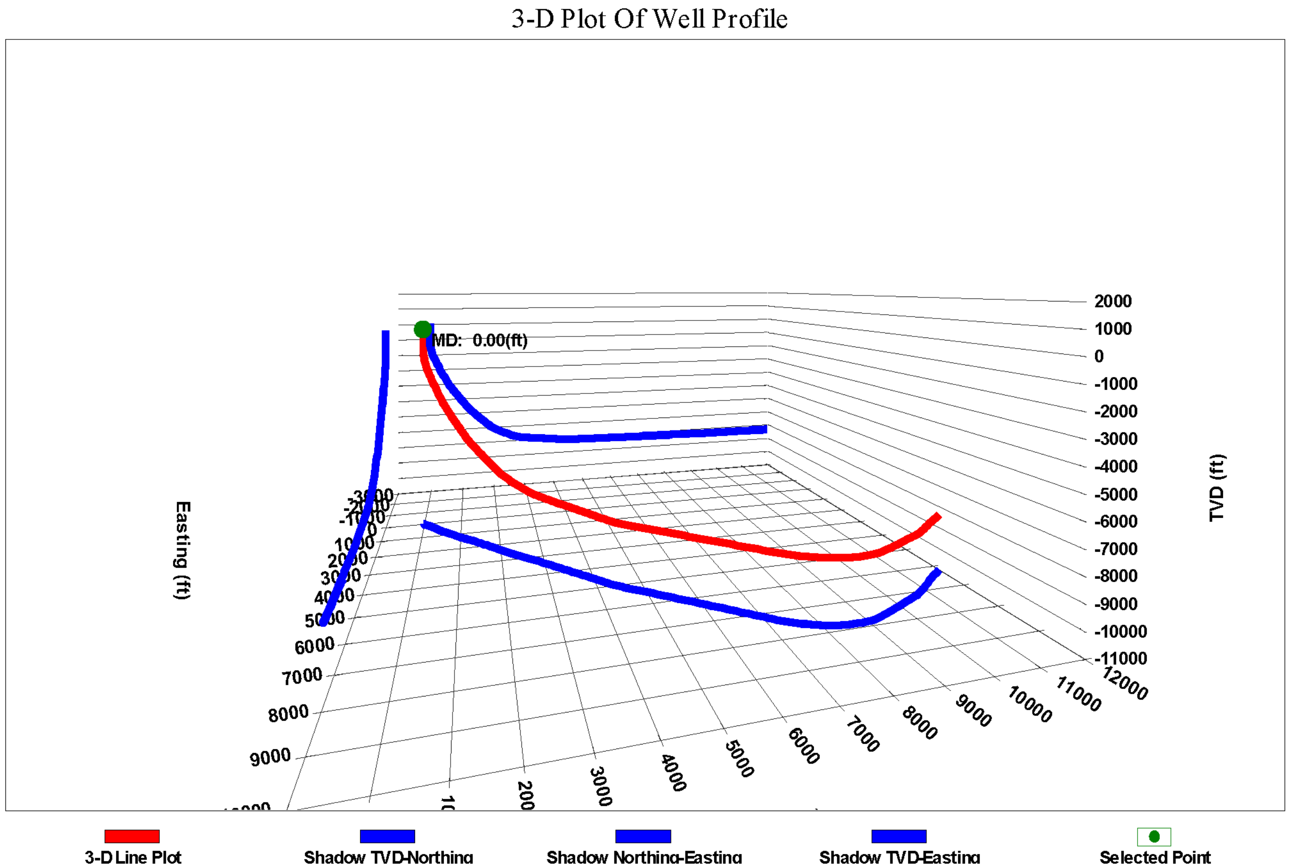

Figure 11—3D plot of well profile for well B

Run 1 – Well B

A Hydraulic Pulsation Tool which is a third party provider with a 0.3-inch orifice, was prepared for stimulation in the complex extended well. At 15,000 ft, treated water (TW) was pumped at 1.0 bpm to activate the ERT. however, unstable CT pressure responses were observed, with surface pressure spiking to 4,500 psi. The pump rate was reduced to 0.8 bpm, stabilizing CT pressure to 4,200 psi, allowing RIH to continue.

At 17,665 ft, a sudden reduction in surface weight from 20,000 lbs to 11,000 lbs indicated potential tool lock-up as can see in Figure 12. Continued attempts to RIH were made, but an additional weight loss to 6,000 lbs at 17,458 ft, under 4,300 psi and 0.8 bpm, confirmed that mechanical advancement had ceased.

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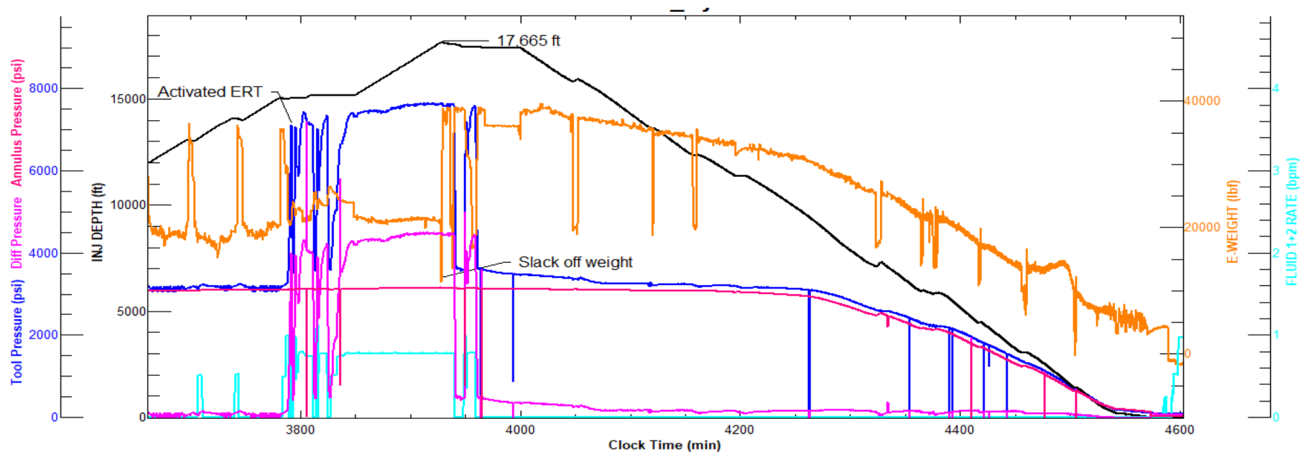

Figure 12—Telemetry-based response during Run 1 showing pressure behavior, CT weight, and depth anomalies around Hydraulic Pulsation Tool activation.

RUN 2- Well B Hydraulic Vibration Tool failure reached TD

At 15,000 ft, the vibration extended-reach tool (ERT) was activated by pumping at 1.2 bpm, resulting in a coiled tubing surface weight of approximately 19,000 lbs and a pump pressure of 1,800 psi. Unfortunately, a loss in telemetry occurred, leading to the loss of communication with the downhole tools, which could no longer be monitored in real-time. However, surface sensors remained functional and allowed the operation to continue.

At 18,350 ft, the pump rate was increased to 1.5 bpm, with surface weight reading approximately 17,000 lbs. Upon reaching 19,045 ft, a sudden drop in weight to –8,000 lbs was observed, indicating potential tool lock-up and unexpected behavior of the extended-reach tool. The decision was made to pull out of hole (POOH) to inspect the BHA. At surface, function testing revealed the circulation sub had opened prematurely, compromising tool function and circulation integrity.

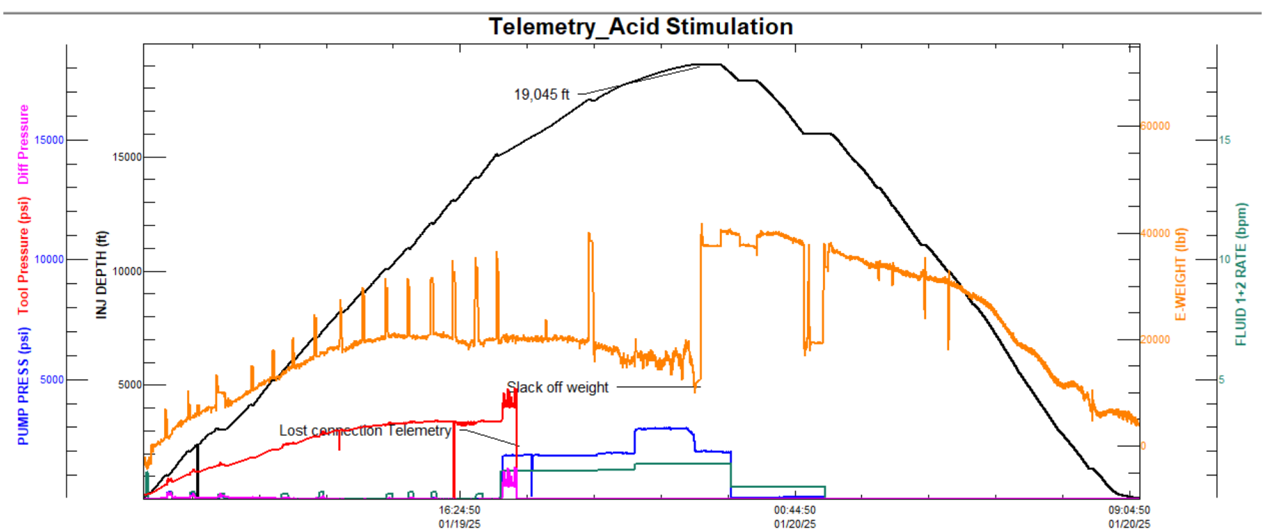

Figure 13—Telemetry trends captured during run 2, highlighting differential pressure, and key operational events such as slack-off and signal loss

Run 3 - Well B Hydraulic Vibration Tool successfully reached TD

Upon reaching 5,760 ft measured depth (MD), a friction-reducing fluid was deployed to mitigate wellbore drag and facilitate smoother coiled tubing (CT) movement. As the CT advanced to 15,000 ft, the pump rate gradually increased to 1.2 barrels per minute (bpm) to hydraulically activate the extended-reach vibration

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tool. Following successful activation, the CT continued to run in hole (RIH) and reached total depth (TD) at 20,600 ft. Throughout the pumping operation, surface parameters remained stable, and downhole differential pressure was maintained between 1,800 and 2,000 psi, indicating proper tool functionality and effective agitation.

At TD, acid stimulation was initiated across 14 stages as show in Figure 14. To ensure effective acid placement in the open hole section—from 20,600 ft to the top of the liner at 17,687 ft—the extended-reach vibration tool was hydraulically isolated. Upon completion of the acid stages, the operation transitioned into the post-flush phase. CT was at 18,265 ft, where a mechanical lock-up occurred, with surface weight readings indicating approximately -7,000 lb. In response, the CT was pulled out of hole (POOH) to the liner top at 17,687 ft, where the remaining post-flush volume was successfully pumped to ensure complete displacement of acid.

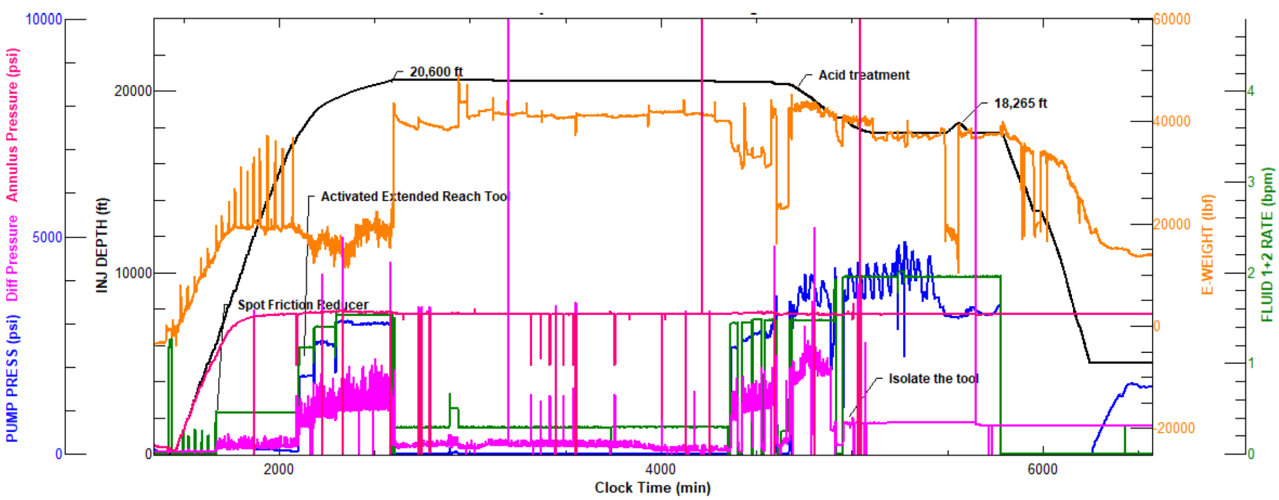

Figure 14—Telemetry profile captured during acid stimulation with hydraulic vibration tool, showing annulus pressure, differential pressure, pump pressure, injected fluid rate, and tool activation events

Hydraulic Multilateral Tool (HMLT)

Hydraulic Multilateral Tool was deployed as part of the coiled tubing bottom-hole assembly (BHA) to enable precise and efficient access to individual laterals in a dual open-hole well completion. The tool featured an integrated casing collar locator (CCL) and downhole pressure telemetry, which provided real-time confirmation of junction location, depth correlation, and lateral orientation. Upon reaching the lateral junction, the tool executed a full circumferential search mode, detecting the lateral branch through a distinct pressure response and automatically aligning for entry. The combination of real-time feedback and hydraulic extension capabilities enabled successful navigation and stimulation of both laterals in a single run, thereby improving intervention efficiency and reducing operational costs (Chishti et al. 2024; Alnasser et al. 2023).

HMLT deployment in complex well architectures, presents significant operational challenges that can be mitigated through the integration of downhole sensors. Traditional sensor less systems rely on mechanical guidance and hydraulic control—typically through surface-modulated pressure and flow sequences—to navigate junctions and access lateral legs. While effective in certain geometries, this passive approach limits the ability to confirm lateral entry, depth correlation, or detect obstructions, often resulting in misruns, leading in extra operational time especially while searching for access into lateral (Lambert 2000; Ravensbergen 2001). In contrast, sensor-enabled systems—utilizing technologies such as, CCL, pressure/temperature sensors—offer real-time feedback on tool location and well conditions. This facilitates active steering and enhances diagnostics (e.g., identifying tool malfunctions, correlating depth), ultimately

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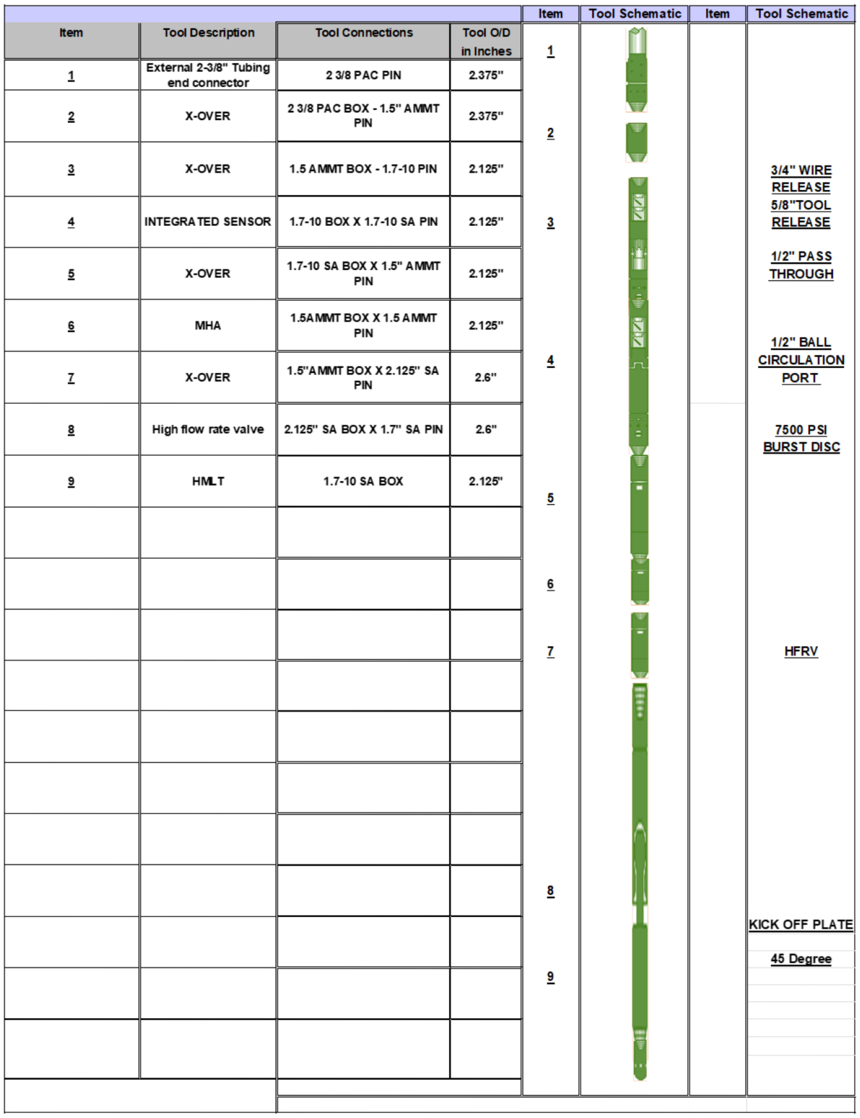
reducing nonproductive time and improving return on investment. To further optimize deployment, HMLT tools integrate sensor packages for real-time identification and utilize telemetry for data transmission.

BHA proposed for HMLT



Item	Tool Description	Tool Connections	Tool O/D in Inches
1	External 2-3/8" Tubing end connector	2 3/8 PAC PIN	2.375"
2	X-OVER	2 3/8 PAC BOX - 1.5" AMMT PIN	2.375"
3	X-OVER	1.5 AMMT BOX - 1.7-10 PIN	2.125"
4	INTEGRATED SENSOR	1.7-10 BOX X 1.7-10 SA PIN	2.125"
5	X-OVER	1.7-10 SA BOX X 1.5" AMMT PIN	2.125"
6	MHA	1.5AMMT BOX X 1.5 AMMT PIN	2.125"
7	X-OVER	1.5"AMMT BOX X 2.125" SA PIN	2.6"
8	High flow rate valve	2.125" SA BOX X 1.7" SA PIN	2.6"
9	HMLT	1.7-10 SA BOX	2.125"

Figure 15—BHA proposed for HMLT

Studie Case

In well C, A dual-lateral matrix acid stimulation run was successfully executed on a dual - seawater power water injector, with the primary objective of mitigating near-wellbore damage and enhancing injectivity across both laterals. The well was completed with two horizontal laterals. Lateral L-0 featured 9 5/8″ casing set at 1,789 ft MD, followed by a 7″ liner to 7,091 ft, and an 8 ½″ open hole section extending to 11,827 ft MD. Lateral L-2 was similarly constructed with 9 5/8″ casing and a 7″ liner to 6,628 ft, with the open hole extending from 6,627 ft to 12,421 ft MD. The well reached a maximum deviation of 90.42°, necessitating the use of coiled tubing (CT) deployed Hydraulic Multilateral tools (HMLT) for selective lateral access and stimulation.

The first run Figure 16 was dedicated to L-2 access and cleanout. Lateral entry was achieved by deploying a HMLT equipped with downhole sensors, including casing collar locator (CCL), pressure sensors, and

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differential pressure transducers. These tools enabled accurate mapping and lateral identification through operational modes including kick-detection, searching, and reset mode. A function test was conducted at 6,000 ft as reference to identify the pressures behavior of the tool. Initial attempt was carried out at 6,640ft, where no drop in pressure while mapping was identified, then in a second attempt to access to L-2 was confirmed at 6,637 ft, based on a drop in pressure while mapping mode as can see in Figure 16, this drop in pressure is an indication that a window was found. A flow-activated circulation valve was then triggered to allow high-rate pumping during the cleanout phase. However, progression was halted due to a mechanical lockup at 9,800 ft MD, despite multiple manipulation attempts.

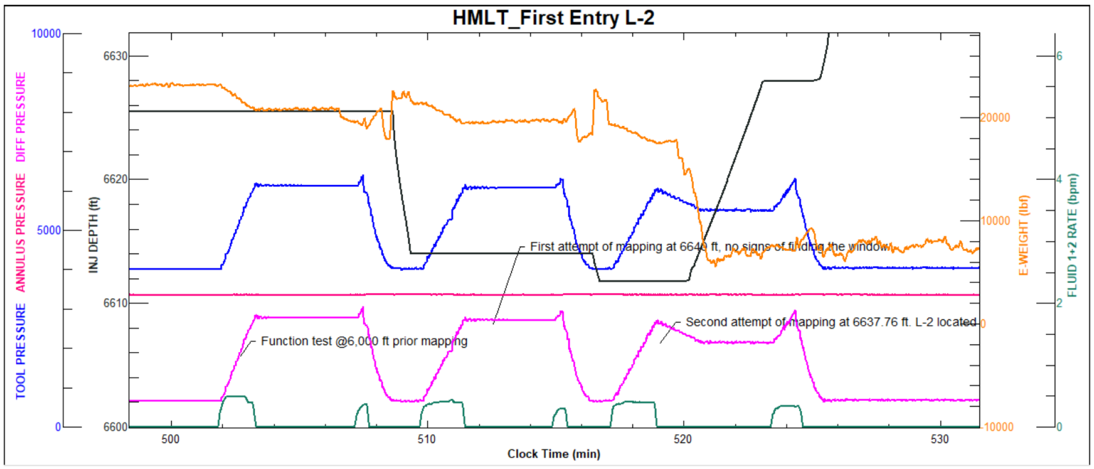

Figure 16—First attempt to Access L-2

Consequently, the CT string was retrieved out the L-2, and a friction reducer pill was deployed across the high-deviation interval from 5,000 to 6,600 ft MD to minimize drag forces and enhance coiled tubing conveyance during subsequent run-in-hole operation

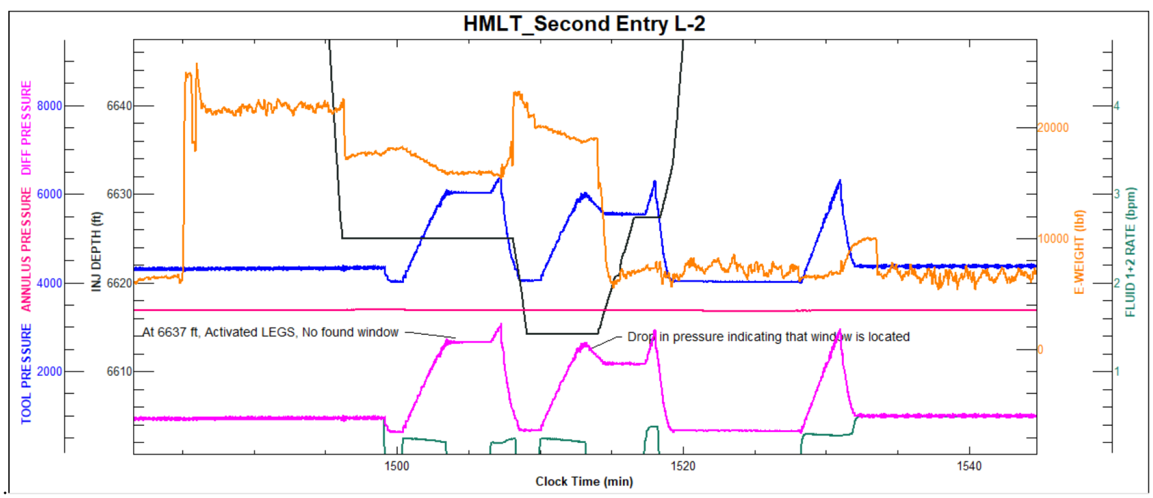

Figure 17—Second successful access to L-2

Subsequently, the window for L-2 was re-entered Figure 17, and the HMLT was redeployed. Successful re-entry was confirmed with CCL and pressure readings, allowing the CT to reach 10,161 ft MD. A decision was made to exit L-2 and access L-0, reaching a total measured depth of 11,791 ft, where acid treatment preparation was initiated. A third and final run was executed to access L-2 again for acid stimulation. Following the same operational parameters and downhole telemetry feedback, the tool once again reached

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10,161 ft MD, where acid and diverter fluids were successfully placed to target damaged intervals and enhance formation contact.

This operation marked the successful completion of three independent L-2 Figure 18 entries within a single CT deployment, optimizing operational efficiency while eliminating the need for additional workover interventions. The effectiveness of the stimulation was confirmed by improved post-job injectivity performance, validating both the tool reliability and the strategic execution methodology applied.

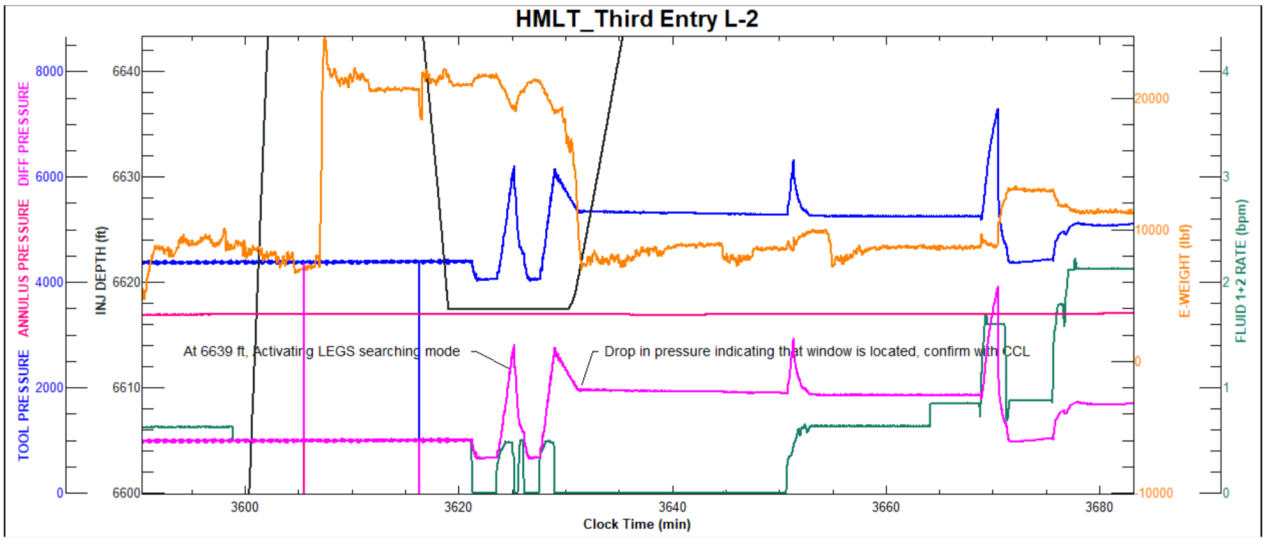

Figure 18—Third L-2 access

Discussion and analysis

In Well A, operational challenge was encountered during the run-in-hole operation. While progressing through the horizontal section, a sudden drop in differential pressure was observed. The operation had proceeded as expected through the liner and into the open hole, with tool activation initiated via pre-flush pumping. However, upon advancing further in the horizontal section, mechanical resistance was encountered, followed by an unexpected pressure drop confirmed by downhole telemetry. This anomaly was identified due to the implementation of real-time telemetry, which enabled early detection of tool behavior. Based on this observation, a decision was made to pull out of hole (POOH) for tool inspection.

The second run use a hydraulic tractor to support extended reach acid stimulation in the horizontal section. While the initial tool response showed signs of proper function, the run faced repeated difficulties, including sudden weight changes and mechanical lockups, These issues led to delays and required pulling out of hole for inspection. Surface testing confirmed a failure in the tractor's gripping mechanism. This experience highlights the operational challenges of running mechanical tools in long horizontal wells and the importance of real-time data, proper tool validation, and contingency planning to reduce the risk of failure.

During the third and last run a vibration extended reach tool was deployed to enhance coiled tubing (CT) reach and enable effective acid stimulation. CT successfully reached 18,900 ft—approximately 95% of the planned open-hole section. Activation of the ERT was confirmed by weight fluctuation analysis and correlated with increasing pump rates. In this case, the availability of downhole telemetry played a critical role, allowing real-time monitoring of tool differential pressure and supporting operational decision-making. Based on telemetry feedback, appropriate pump rate adjustments were made, enabling the CT to meet the client's target depth requirements.

In Well B, three coiled tubing intervention runs were conducted, each contributing to a deeper understanding of tool efficiency and the challenges faced in long horizontal interventions

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Run 1 utilized a hydraulic pulsation tool, The tool's pressure response was highly sensitive to pump rate, and its behavior showed pressure spikes and signs of vibration failure. These patterns were consistent with failure modes observed in similar extended-reach campaigns (Mullen et al., 2014). A sudden weight loss at depth, combined with pressure instability, suggested inadequate axial force transfer, likely limiting tool effectiveness in horizontal section.

Run 2 experienced a loss of communication with the downhole tool, resulting in an unmonitored run with no access to real-time telemetry. Early signs of operational issues—including slack-off in weight and minor surface pump pressure fluctuations—were observed but could not be accurately interpreted without downhole data. These indicators hinted at downhole restrictions or tool malfunction, but the absence of telemetry delayed diagnosis and decision-making. Ultimately, a burst disk was found to have activated prematurely, further confirming tool failure. This run emphasizes the critical role of telemetry in detecting downhole conditions and enabling timely, informed responses.

Run 3 implemented a hydraulic vibration tool designed to generate axial vibrations and reduce friction along the CT string, particularly in high-angle and horizontal sections. The tool allowed continued run-in-hole (RIH) to total depth (TD) at 20,600 ft. Real-time monitoring of tool response improved control over pump rates and treatment placement. The successful outcome of this run highlights the value of proactive drag management, dynamic circulation strategies, and tool-assisted reach enhancement. The combination of effective friction-reducing fluids, reliable tool activation, and continuous telemetry contributed to a high-efficiency stimulation treatment in a complex horizontal wellbore

In well C, an acid stimulation operation in the dual-lateral seawater power water injector well was completed successfully due in large part to the deployment of advanced coiled tubing (CT) telemetry and downhole sensor technologies merged with a Reliable Hydraulic Multilateral tool. While frictional resistance and chemical considerations were notable, the precision and responsiveness provided by real-time downhole data were the most critical enablers of operational success.

Initial access to L-2 Figure 16, the integration of a real-time telemetry-enabled bottom hole assembly (BHA) equipped with casing collar locator (CCL), tool string pressure sensors, and differential pressure monitoring allowed for accurate mapping and lateral identification during each entry attempt. The use of multi-mode operation—including kick-detection, search, and reset functions—enabled dynamic tool behavior adjustment in response to wellbore conditions.

These sensors were particularly valuable during lateral re-entries. The high-resolution pressure data not only confirmed correct positioning within the target lateral but also provided early indication of tool engagement. This capability reduced the need for repeated pull-out-of-hole (POOH) cycles and ensured that all three successful entries into L-2 were completed within a single CT deployment. The flow-activated circulation valve, also monitored through pressure telemetry, allowed high-rate circulation during cleanout operations while maintaining control of tool activation sequences.

Lessons Learned

- The hydraulic pulsation tool exhibited high sensitivity to pump rate fluctuations, resulting in pressure instability and premature lock-up. This underscores the importance of precise flow control when deploying vibratory tools in extended-reach environments
- Sudden weight loss at depth can serve as an early indicator of lock-up, requiring immediate contingency planning.
- Burst disk failure during stimulation highlighted the need for reinforced pressure integrity within bottomhole assemblies (BHAs), particularly when deployed alongside high-vibration tools such as hydraulic vibration and pulsation devices. The dynamic loads and transient pressure fluctuations generated during tool activation can increase stress on critical components like rupture disks.

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- The integration of real-time telemetry significantly enhanced operational decision-making, enabling accurate depth tracking, tool validation, and early detection of frictional lockups, which collectively minimized non-productive time (NPT)
- The gripper pin shearing incident in the hydraulic tractor tool highlighted vulnerabilities in mechanical components subjected to dynamic axial and torsional loads. These stress concentrations emphasize the need for reinforced interfaces and predictive stress modeling to enhance tool durability and reduce the risk of failure during critical intervention sequences.

Conclusions

- Real-time downhole telemetry proved essential for early identification of abnormal pressure behavior, allowing on time decisions to mitigate tool failure. While mechanical and hydraulic tools encountered functional limitations—including mechanical lockups and premature burst disk activation- This progression highlights the value of real-time diagnostics, dynamic pump rate control, and tool optimization in managing weights lockup and extending operational reach
- In Well A, mechanical resistance and a sudden pressure drop led to the decision to pull out of hole (POOH). Real-time telemetry confirmed unusual tool behavior, allowing the team to act early and avoid further issues.
- In Well A, Decision to adjust pump rates based on real-time differential pressure and weight fluctuation data. Enabled CT to reach 18,900 ft and meet stimulation objectives.
- In Well B, three runs revealed distinct tool behavior and failure mechanisms. Hydraulic pulsation tools showed sensitivity to pump rate and insufficient axial transfer, while the absence of telemetry in the second run led to delayed failure detection and unplanned tool activation. The third run, utilizing a hydraulic vibration tool with active telemetry, successfully reached total depth and enabled high-efficiency stimulation. These results reinforce the importance of merging advanced tools with continuous data acquisition for effective extended-reach interventions.
- In a HMLT, Real-time downhole telemetry was the single most critical factor in ensuring precise lateral access, tool function verification, and acid placement control during stimulation of both L-0 and L-2.
- The use of integrated pressure and differential sensors, combined with CCL-based depth correlation, enabled highly accurate lateral mapping and navigation, even under challenging wellbore conditions and friction-related constraints.
- The combination of telemetry-enabled tool control and advanced lateral access workflows allowed three successful L-2 entries and stimulation placement to be executed within a single intervention, significantly improving operational efficiency and reducing costs.
- Accurate depth tracking using CCL sensors in HMLT applications plays an important role in reducing operational time. It allows access to laterals with fewer attempts compared to depth tracking based solely on surface sensors
- Dynamic telemetry feedback minimized operational risks by allowing early detection of tool lock-up, formation response, and fluid placement performance, reducing NPT and eliminating the need for additional workover equipment.

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Nomenclature

Symbol = Description

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CT = Coiled Tubing
MD = Measured Depth, ft
TVD = True Vertical Depth, ft.
BHA = Bottomhole Assembly.
ESP = Electric Submersible Pump.
Bpm = Barrels per minute.
Psi = Pounds per square inch.
NPT = Non-Productive Time.
RIH = Run In Hole
POOH = Pull Out Of Hole.
TD = Total Depth.
ERT = Extended Reach Tool
H_2S = Hydrogen Sulfide
HMLT = Hydraulic Multilateral Tool.
CCL = Casing Collar Locator

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